

AI ASSISTED CODING

ASSIGNMENT-10.1

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Batch-40

Task Description #1 – Syntax and Logic Errors

Task: Use AI to identify and fix syntax and logic errors in a faulty Python script.

Sample Input Code:

```
# Calculate average score of a student
```

```
def calc_average(marks):
```

```
total = 0
```

```
for m in marks:
```

```
total += m
```

```
average = total / len(marks)
```

```
return avrage # Typo here
```

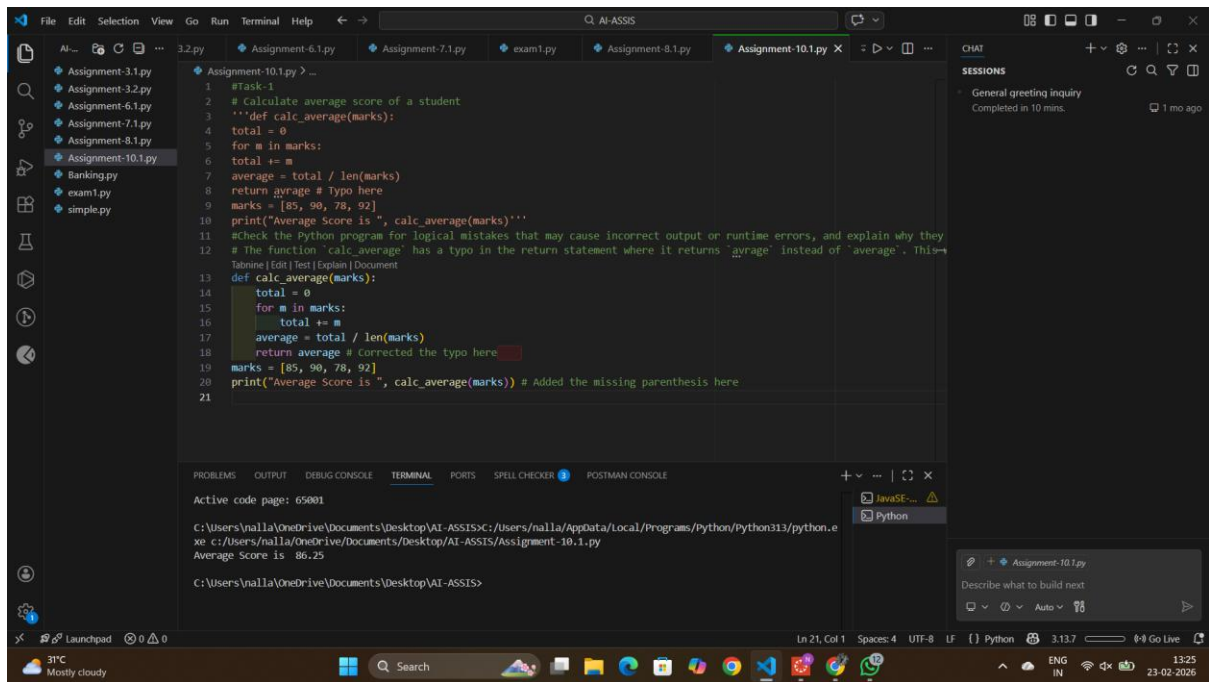
```
marks = [85, 90, 78, 92]
```

```
print("Average Score is ", calc_average(marks))
```

Expected Output:

- Corrected and runnable Python code with explanations of the fixes.

Code and Output:



Analysis:

- The AI detects indentation mistakes, missing brackets, and spelling errors in variables
- It identifies logic flaws such as incorrect variable usage or faulty return values
- The AI corrects the code and produces a runnable program with explanations

Task Description #2 – PEP 8 Compliance

Task: Use AI to refactor Python code to follow PEP 8 style guidelines.

Sample Input Code:

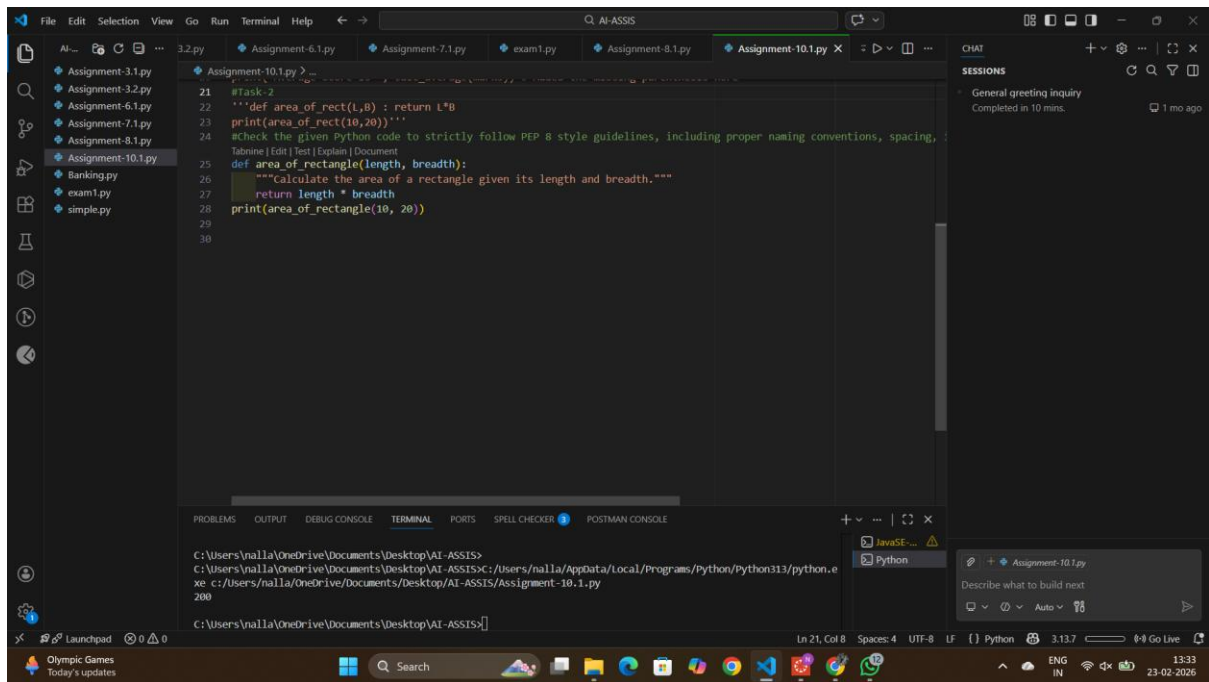
```
def area_of_rect(L,B) : return L*B
```

```
print(area_of_rect(10,20))
```

Expected Output:

- Well-formatted PEP 8-compliant Python code.

Code and Output:



Analysis:

- The AI enforces proper naming conventions and spacing rules
- It restructures code layout for better readability and consistency
- The program remains functionally unchanged but becomes standard-compliant

Task Description #3 – Readability Enhancement

Task: Use AI to make code more readable without changing its logic.

Sample Input Code:

```
def c(x,y):
    return x*y/100

a=200

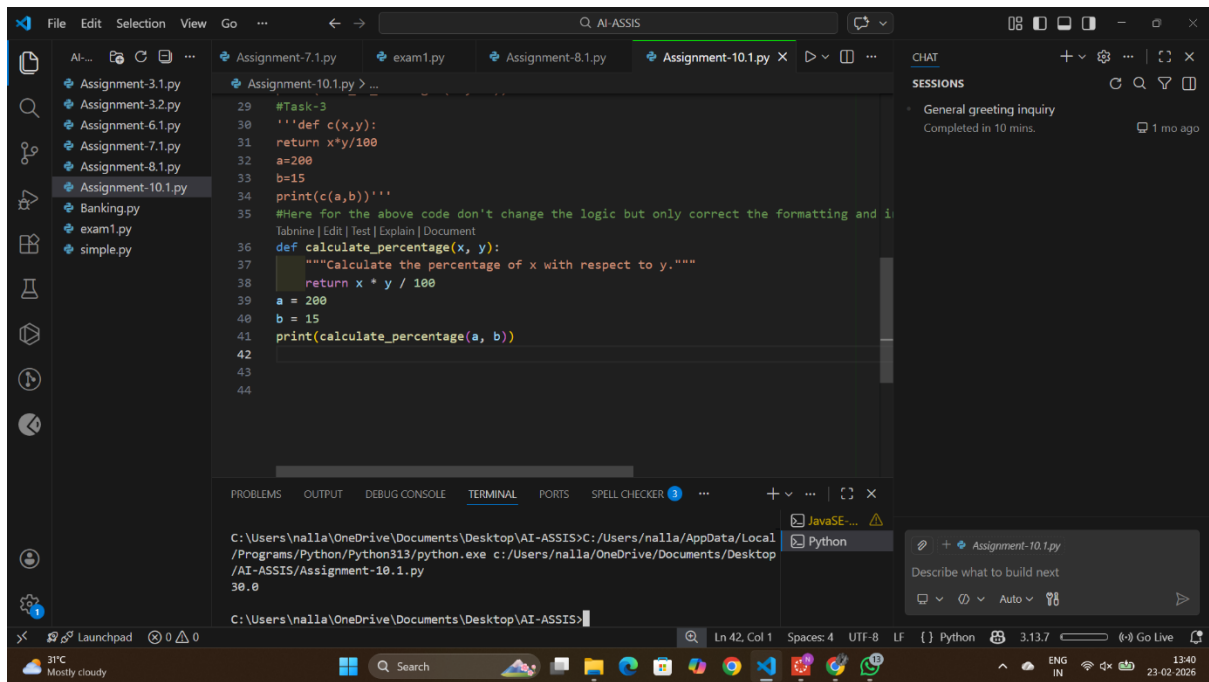
b=15

print(c(a,b))
```

Expected Output:

- Python code with descriptive variable names, inline comments, and clear formatting.

Code and Output:



Analysis:

- The AI replaces unclear variable names with descriptive ones
- It adds meaningful comments explaining each step
- The code is neatly formatted for easy understanding

Task Description #4 – Refactoring for Maintainability

Task: Use AI to break repetitive or long code into reusable functions.

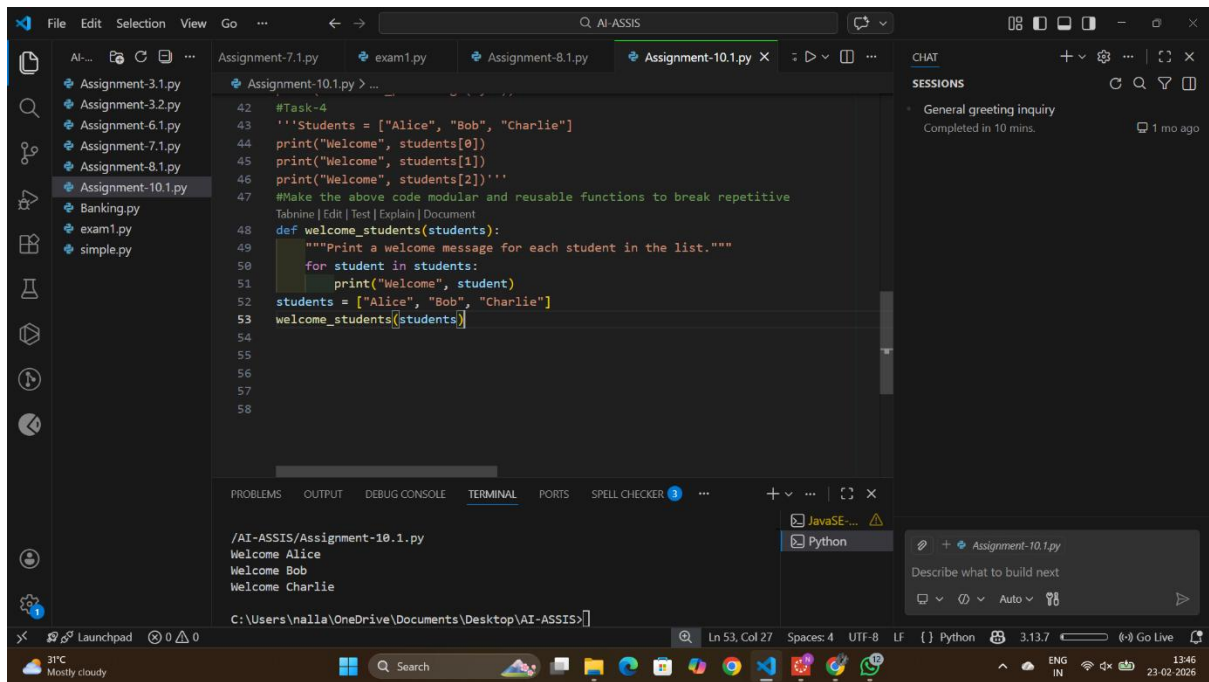
Sample Input Code:

```
students = ["Alice", "Bob", "Charlie"]  
  
print("Welcome", students[0])  
  
print("Welcome", students[1])  
  
print("Welcome", students[2])
```

Expected Output:

- Modular code with reusable functions.

Code and Output:



Analysis:

- The AI identifies repeated code blocks
- It converts them into reusable functions
- This makes the program easier to update and maintain

Task Description #5 – Performance Optimization

Task: Use AI to make the code run faster.

Sample Input Code:

Find squares of numbers

```
nums = [i for i in range(1,1000000)]
```

```
squares = []
```

for n in nums:

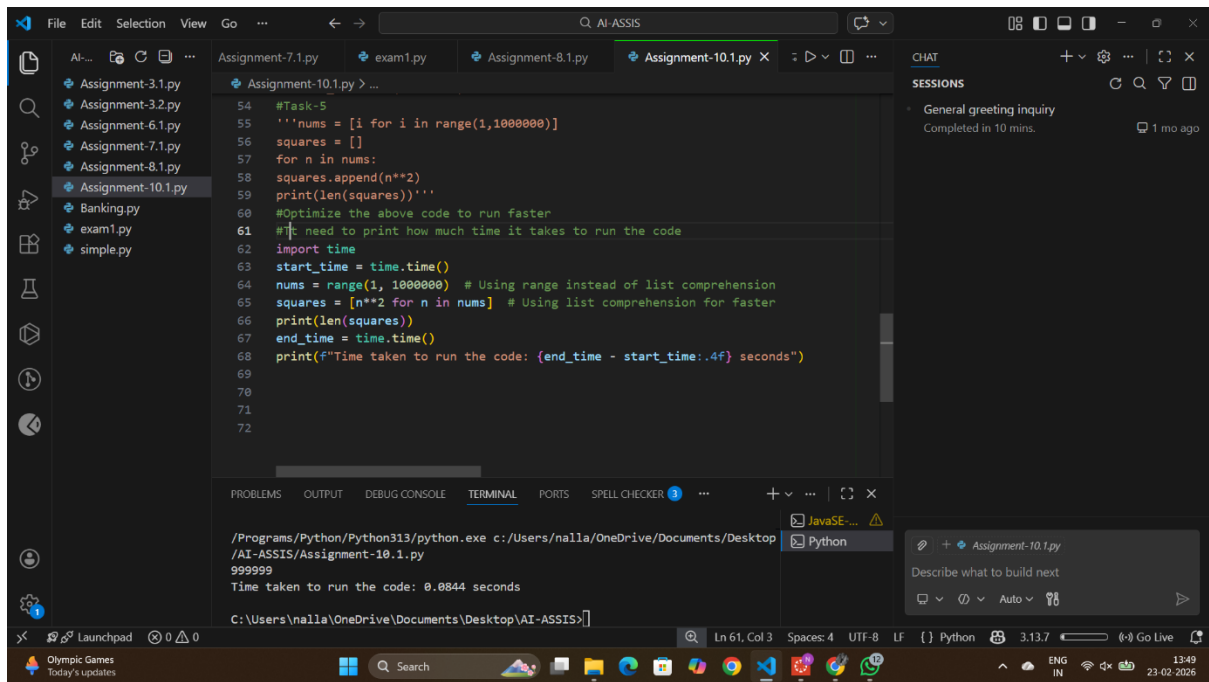
```
squares.append(n**2)
```

```
print(len(squares))
```

Expected Output:

- Optimized code using list comprehensions or vectorized operations.

Code and Output:



Analysis:

- The AI removes unnecessary loops and intermediate lists
- It uses faster techniques like list comprehensions
- The code executes quicker with lower memory usage

Task Description #6 – Complexity Reduction

Task: Use AI to simplify overly complex logic.

Sample Input Code:

```
def grade(score):
```

```
if score >= 90:
```

```
    return "A"
```

```
else:
```

```
if score >= 80:
```

```
    return "B"
```

```
else:
```

```
if score >= 70:
```

```
    return "C"
```

```
else:
```

```
if score >= 60:
```

```
    return "D"
```

else:

return "F"

Expected Output:

- Cleaner logic using elif or dictionary mapping.

Code and Output:

```
70 #Task-6
71 '''def grade(score):
72     if score >= 90:
73         return "A"
74     else:
75         if score >= 80:
76             return "B"
77         else:
78             if score >= 70:
79                 return "C"
80             else:
81                 if score >= 60:
82                     return "D"
83                 else:
84                     return "F"'''
85 #The above code is a complex one make it simpler by using elif statements
86 def grade(score):
87     if score >= 90:
88         return "A"
89     elif score >= 80:
90         return "B"
91     elif score >= 70:
92         return "C"
93     elif score >= 60:
94         return "D"
95     else:
96         return "F"
97 #take one sample input and print the grade
```

Terminal Output:

```
c:\Users\nalla\OneDrive\Documents\Desktop\AI-ASSIS>python
.exe c:\Users\nalla\OneDrive\Documents\Desktop\AI-ASSIS\Assignment-10.1.py
The grade for score 85 is: B
```

```
80 else:
81     if score >= 60:
82         return "D"
83     else:
84         return "F"'''
85 #The above code is a complex one make it simpler by using elif statements
86 def grade(score):
87     if score >= 90:
88         return "A"
89     elif score >= 80:
90         return "B"
91     elif score >= 70:
92         return "C"
93     elif score >= 60:
94         return "D"
95     else:
96         return "F"
97 #take one sample input and print the grade
98 score = 85
99 print(f"The grade for score {score} is: {grade(score)}")
```

Terminal Output:

```
c:\Users\nalla\OneDrive\Documents\Desktop\AI-ASSIS>python
.exe c:\Users\nalla\OneDrive\Documents\Desktop\AI-ASSIS\Assignment-10.1.py
The grade for score 85 is: B
```

Analysis:

- The AI simplifies nested conditional statements
- It replaces complex logic with cleaner structures like `elif`
- The result is shorter, clearer, and easier-to-debug code